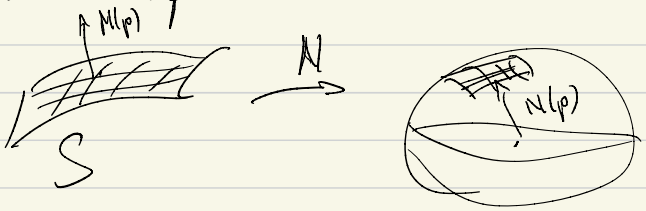
The Gauss map

Suppose $S \subset \mathbb{R}^3$ is a regular surface with unit normal vector N

Let $\mathbb{S}^2$ denote the unit sphere in $\mathbb{R}^3$

The Gauss map is $N : S \longrightarrow \mathbb{S}^2$



As a plane in $\mathbb{R}^3$, $T_{N(p)}\mathbb{S}^2$ = plane $\perp$ to $N(p)$

$= T_pS$

Hence we regard the differential as a map

$$dN_p : T_pS \longrightarrow T_pS$$

Prop. $\mathrm{II}(X,Y) = -g(dN(X), Y)$ for $X, Y \in T_pS$

Proof: Use local coordin. $(u^1, u^2) \longrightarrow \partial_1, \partial_2$

Writing N as $N(u^1, u^2)$ gives

$dN(\partial_j) = N_j$ (meaning $\frac{\partial}{\partial u^j} N$)

Thus, $g(dN(\partial_i), \partial_j) = N_i \cdot \sigma_j$

Since $N \cdot \sigma_j = 0$, $N_i \cdot \sigma_j + N \cdot \sigma_{ij} = 0$

$\therefore g(dN(\partial_i), \partial_j) = -N \cdot \sigma_{ij}$

$= -\mathrm{II}(\partial_i, \partial_j)$ ✓

define = the Gaussian curvature of S at p is

$$K(p) := -\det(dN_p)$$

the mean curvature is

$$H(p) := -\frac{1}{2}\operatorname{tr}(dN_p)$$

(Note that H depends $\pm$ on N, but K is independent)

How to compute?

In coordinates, we can write $-dN_p$ as a matrix

$$-dN_p(\partial_i) = W_i^j \partial_j \qquad (W \text{ for "Weingarten map"})$$

then $K = \det W$, $H = \frac{1}{2} W_j^j$

Let $\mathrm{II}_{ij} := \mathrm{II}(\partial_i, \partial_j) = N \cdot \sigma_{ij}$

By the Prop. $\mathrm{II}_{ij} = g(W_i^k \partial_k, \partial_j)$

$$= W_i^k g_{kj}$$

Convention: $[g^{ij}] :=$ inverse matrix to $[g_{jk}]$

$$\Rightarrow g^{ij} g_{jk} = \delta_k^i \qquad (\text{identity matrix})$$

This gives $W_i^k = \mathrm{II}_{ij} g^{jk}$

$$\therefore \boxed{K = \frac{\det \mathrm{II}}{\det g}}$$

Since the trace does not factor,

$$H = \frac{1}{2} \mathrm{II}_{ij} g^{ji}$$

Examples

1) catenoid $\sigma(z,\theta) = (\cosh z \cos\theta, \cosh z \sin\theta, z)$

$\sigma_1 = (\sinh z \cos\theta, \ , 1)$
$\sigma_2 = (-\cosh z \sin\theta, \ , 0)$

$N = \dfrac{(-\cos\theta, -\sin\theta, \sinh z)}{\cosh z}$

$[g_{ij}] = \begin{pmatrix} \cosh^2 z & 0 \\ 0 & \cosh^2 z \end{pmatrix}$ $\quad [g^{jk}] = \begin{pmatrix} 1/\cosh^2 z & 0 \\ 0 & 1/\cosh^2 z \end{pmatrix}$

$\sigma_{11} = (\cosh z \cos\theta, --, 0)$
$\sigma_{12} = (-\sinh z \sin\theta, \ , 0)$
$\sigma_{22} = (-\cosh z \cos\theta, --, 1)$

$[\mathrm{II}_{ij}] = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow [W_i^j] = \begin{pmatrix} -\frac{1}{\cosh^2 z} & 0 \\ 0 & +1/\cosh^2 z \end{pmatrix}$

$K = -\dfrac{1}{\cosh^4 z}$, $H = 0$

2) helicoid $\sigma(s,t) = (s\cos t, s\sin t, t)$

$\sigma_1 = (\cos t, \sin t, 0)$
$\sigma_2 = (-s\sin t, s\cos t, 1)$

$N = \dfrac{(\sin t, -\cos t, s)}{\sqrt{1+s^2}}$

$ds^2 = ds^2 + (1+s^2)dt^2 \Longrightarrow [g^{jk}] = \begin{pmatrix} 1 & 0 \\ 0 & 1/1+s^2 \end{pmatrix}$

$\sigma_{11} = 0$
$\sigma_{12} = (-\sin t, \cos t, 0)$
$\sigma_{22} = (-s\cos t, -s\sin t, 0)$

$\Rightarrow [\mathrm{II}_{ij}] = \begin{pmatrix} 0 & 1/\sqrt{1+s^2} \\ 1/\sqrt{1+s^2} & 0 \end{pmatrix}$

$[W_j^k] = \begin{pmatrix} 0 & (1+s^2)^{-1/2} \\ (1+s^2)^{-3/2} & 0 \end{pmatrix}$

$K = -(1+s^2)^{-2}$, $H = 0$

(Recall the local isometry was $s = \sinh z$, $t = \theta$)